

Verification Revamp

Summary

RPA

Hypothesis

User Problem

Business Problem

Scope

Figma

Use Cases

Edge cases

Add

TBD

Future scope

Misc

Summary

- The document outlines the approach for enhancing the PAN, KYC, UPI, Bank & Email Verification journeys.

RPA

- As a user, I should be able to withdraw seamlessly and faster

Hypothesis

- Reduce first withdrawal 90th percentile TAT from ~35 minutes to under 2 minutes

User Problem

Problems with first time withdrawal

- As a user, I feel as if Dream11 is enforcing a lot of steps, purposely adding friction to not allow me to place my first withdrawal

Problems with Email Verification

- As a user, I feel email isn't required to withdraw my money

Problems with PAN Verification

- As a user, I have to fill-up a form of my PAN details & upload an image as well, which is sort of redundant

- I have to go back and forth to check any error messages as we don't send comms to the user.

Problems with the UPI Verification Flow

- As a user, I don't remember my UPI VPA address

Problems with the Bank Verification Flow

- A real friction for me to verify my bank due to the documents that need to be uploaded.
As a user, I this isn't something that I usually carry with me or have it quite handy.

Business Problem

Problems with first time withdrawal

- 90th percentile TAT to place my first withdrawal is 35 minutes

Problems with Email Verification

- Additional step increase the TAT for first withdrawal.
- Capture user's email when the intent is high for sending comms.

Problems with PAN Verification

- Is the PAN verification required for net losers ?

Problems with the UPI Verification Flow

- Increase exposure to UPI

Problems with the Bank Verification Flow

- 90th percentile TAT is ~6 minutes

Scope

- Withdrawal flow
 - Initiate PAN, Bank, UPI & Email verification from the withdrawal flow
 - Nudging user to select the better option based on 90th percentile TAT for Bank & UPI
 - Ability to unlink Bank, UPI or Email
 - Add KYC verification at withdrawal
- Changes in Email Verification flow
 - Skip email verification flow for user's first withdrawal
- Changes in PAN Verification
 - Integrate new partners

- Sync flow - show success / errors on request submit
- PAN verification validation
- PAN verification options (TBD)
- Changes in UPI Verification flow
 - New flow based on Mobile Number - Tap yes / no to verify VPA
- Changes in Bank Verification flow
 - Sync flow - show success / errors on request submit
 - Only ask user for document upload option after X failed attempts (configurable)
 - Bank account dedupe
- Improving the My KYC details Page
 - Add KYC verification option on the Verify KYC page

Figma

Use Cases

Use Case	Description
1	Changes in Withdrawal flow
1.1	Initiate PAN, Bank, UPI & Email verification from the withdrawal flow
	Bank verification why TAT is high ? todo
1.2	Nudging user to select the better option based on 90%ile TAT for Bank & UPI • For example, Bank TAT - 5 minutes. UPI TAT - 7 seconds. • If user has selected Bank for withdrawing, nudge the user to either opt for UPI or say that it will take 5 minutes to receive the withdrawal • Manual withdrawal or max withdrawals (above 2 lacs) - may take upto 3 days.
1.3	Add KYC verification at withdrawal

	• If a user's KYC is aadhaar lite verified or reKYC status or min-KYC pending get the user to verify once again with either OKYC, OCR or DigiLocker.
1.4	Ability to unlink Bank, UPI or Email • There should be a way for the user to be able to unlink their bank, UPI or Email • Bank & UPI unlinking already exists
2	Changes in Email Verification flow
2.1	If user has withdrawn X-1 times & email doesn't exist, skip email verification flow • If the user has withdrawn X-1 times withdrawal transaction and his email address does not exist in the system, the email verification flow will be skipped.
2.2	If user has withdrawn X times & email doesn't exist, open email verification flow • If the user has withdrawn X times and his email address does not exist in the system, the email verification flow will be triggered
2.3	If email exists & user is withdrawing, do nothing • If the user's email address exists in the system & user is withdrawing, the email verification flow will be skipped.
2.4	Email verification flow Bottom sheet with the email verification • Email Verification Form : ◦ Email input field (email format should be validated as the user completes entering the "@domain.com") ◦ Tappable pills (for top email domain suggestions) ◦ Submit button • Once the user taps on submit button, BE will send an OTP to the entered email & FE will show an OTP screen for the user to enter OTP. • OTP copying & pasting ability ◦ In the OTP email, user should be able to copy the OTP ◦ In the OTP screen, user should be able to paste the OTP

	• Success ◦ Entered OTP is validated & email isn't verified on another D11 account • Errors ◦ Wrong OTP entered ◦ Entered OTP is validated but email is already verified on another D11 account. Now user will have to restart the email verification flow • Comms ◦ PN / WhatsApp / Email / App inbox / Toast message : You've successfully verified your email
3	Changes in PAN Verification
3.1	Integrate new partners - Idfy & / or NSDL direct
3.2	Sync flow - show success / errors on request submit
3.3	PAN verification validation (behind feature flag) • if PAN not verified & user is net loser as on date, skip PAN verification flow • if PAN not verified & user is net winner as on date, open PAN verification flow • if PAN verified, skip PAN verification flow
3.4	PAN verification flow options
3.4.1	Option 1 - only user inputs • Input ◦ User entered PAN number, User entered name, User entered DOB • Flow ◦ NSDL call with PAN number, Name, DOB • NSDL response ◦ If invalid PAN, show error ◦ If valid PAN, Name not matching, DOB not matching, show error ◦ If valid PAN, Name not matching, DOB matching, show error ◦ If valid PAN, Name matching, DOB not matching, show error ◦ If valid PAN, Name matching, DOB matching, verify PAN • Handling of false negative ◦ Name not matching

	▪ CX touchpoint - Create a ticket if name entered is as per PAN card with PAN image ◦ DOB not matching ▪ CX touchpoint - Create a ticket if DOB entered is as per PAN card with PAN image • Issues with this flow ◦ PAN image isn't captured
3.5	Show KYC name if KYC is verified
3.6	Success & Error states • Error ◦ If PAN number is invalid ▪ Highlight PAN number field & show error ▪ Comms - PN / WhatsApp / Email : Please enter valid PAN details ◦ If Name is not matching ▪ Highlight Name field & show error ▪ Comms - PN / WhatsApp / Email : Please enter name exactly as on PAN card ◦ If DOB is not matching : ▪ Highlight DOB field & show error ▪ Comms - PN / WhatsApp / Email : Please enter DOB exactly as on PAN card • Success - If PAN is verified & its not verified against any other account ◦ Comms - PN / WhatsApp / Email : You've successfully verified your PAN details
3.7 TODO	PAN Enumeration: Check for PAN dedupe after we get NSDL response A malicious user can keep sending the requests to the service by changing the characters of the PAN and understand if this PAN number is on Dream11 or not. Currently we get this error : "This PAN is already verified against a different user" check with tech - rate limit 10 requests

3.8	PAN unlinking count • We need to log whenever a PAN is verified & unlinked against a User ID. • If a PAN is unlinked X times, same PAN shouldn't be allowed to be auto verified & should be flown to manual queue
3.9	PAN verification for depositors (behind feature flag) • If the life time cumulative deposits for a user is more than Rs. 50k (configurable), user has to verify his PAN.
3.10	PAN dedupe Error If PAN verified on system - show last 2 digits of mobile number of user id where this pan is verified
4	Changes in UPI Verification flow
4.1	New flow based on Mobile Number This flow needs to shown only when we have a name match (internal algo) between (PAN or KYC) with VPA name & the name match response is VERIFIED. • If PAN is verified, PAN Name • If min-KYC is verified, KYC Name Flow : • User's registered mobile number → Get VPA & Name. Multiple VPA could be registered on the same mobile number. • Name match with PAN name, If name matching, ◦ Tap yes to verify VPA ◦ Tap no to reject VPA Success : • Comms - PN / WhatsApp / Email : You've successfully verified your UPI VPA Error : • Comms - PN / WhatsApp / Email : Please verify a valid UPI VPA
4.2	Show Name on UPI verification screen • Show PAN Name, if PAN is verified

	• Show KYC Name, if min-KYC is verified
4.3	UPI name match • If PAN is verified, UPI name & PAN Name needs to name matched • If min-KYC is verified, UPI name & KYC Name needs to name matched
4.4	UPI VPA Enumeration: Check for UPI VPA dedupe after we get name response A malicious user can keep sending the requests to the service by changing the characters of the UPI VPA and understand if this VPA is on Dream11 or not.
5	Changes in Bank Verification flow
5.1.a	Sync flow - show success / errors on request submit
5.1.b	Async flow - show success / errors on request submit
5.2	If user's bank verification is first / X attempts (configurable), skip document upload option • Do not ask the user to upload a doc for the first / X attempts (configurable) • Ask for document after x attempts if the verification is still not verified
5.3	Option to paste the bank A/C number
5.4	Bank account dedupe • Unique key - Account number & first four digits of IFSC • Current dedupe - Account number & IFSC
5.5	Show Name on Bank verification screen • Show PAN Name, if PAN is verified • Show KYC Name, if min-KYC is verified
5.6	Bank name match • If PAN is verified, Bank name & PAN Name needs to name matched • If min-KYC is verified, Bank name & KYC Name needs to name matched
5.6	Bank unlinking count • We need to log whenever a Bank is verified & unlinked against a User ID.

5.7	Bank Account Enumeration: Check for Bank Account dedupe after we get penny drop response A malicious user can keep sending the requests to the service by changing the characters of the Bank account number and understand if this Bank account number is on Dream11 or not.
5.8	Bank verification comms Success : • Comms - PN / WhatsApp / Email : You've successfully verified your Bank Account Error : • Comms - PN / WhatsApp / Email : Please verify a valid Bank Account
5.9	Bank verification rate limit Allow only 3 bank verification requests in a day
6	KYC verification
6.1	Improving the My KYC details Page • Add KYC verification option on the Verify KYC page
6.2	KYC unlinking count • We need to log whenever a KYC is verified & unlinked against a User ID. • If a KYC is unlinked X times, same KYC shouldn't be allowed to be auto verified & should be flown to manual queue
6.3	Dedupe for all KYC modes Users can create multiple accounts using different KYC modes like Aadhaar, Driving License & Voter ID • Initiating min-KYC verification via Aadhaar (OCR/OKYC/DigiLocker) / Voter ID / Driving License / Passport ◦ Search for name (fuzzy name match), dob & state in Aadhaar, Voter ID, Driving License, Passport DB ◦ If any existing KYC entry matches with name, dob & state, we'll reject the verification request & create a KYC dedupe error log entry with user id, name, dob & state

	◦ If name, dob & state doesn't match with any existing KYC entry, we'll approve the verification request
6.4	Error TODO If KYC verified on system - show last 4 digits on mobile number
6.4	Aadhaar Verification via OCR • Ensure to mask the first 8 digits of the Aadhaar number, QR code • Ensure that the Aadhaar number on front image and back image is matching
6.5	Aadhaar number validation Once the user enters a aadhaar number, we will validate it on backend and only • If its a valid aadhaar number, we'll forward the request to the partner. • If its not a valid aadhaar number, we'll display invalid aadhaar number error on FE. Logic :  Aadhaar validity logic.pdf
6.6	KYC verification comms Success : • Comms - PN / WhatsApp / Email : You've successfully verified your KYC Error : • Comms - PN / WhatsApp / Email : Please verify a valid KYC
6.7	RPA of Min-KYC Verify user's following info from RBI prescribed OVD : • Name • Age • State (for Legal & GST both) • Photo
6.8	Min-KYC flow can be invoked at the time of & is configurable via a Feature Flag • Add Cash - should complete min-KYC before adding cash

	• Paid contest join - should complete min-KYC before joining paid contest • Withdrawal - should complete min-KYC before withdrawal • Deeplink Users qualifying for min-KYC • Aadhaar lite verified users • reKYC users • min-KYC not verified users • Users with missing info (Name / State / DOB / Photo)
6.9	Min-KYC methods
6.9.a	Option 1 - DigiLocker / OKYC (Flow remains as it is)
6.9.b	Option 2 - DL + DoB (New Flow) For DoB, we will use the existing calendar format of the OS Dedupe logic for DL • No other DL with the same number should be present in our database
6.9.c	Option 3 - OCR - Upload Image Supported Options - Aadhaar, DL via - Take a Photo, Upload Photo from Gallery, Upload Document Series of steps 1. Tampering Check (Integration doc - <todo>) 2. Doc Classification (already live) 3. OCR info extraction (already live) 4. Verify info from Source a. For Aadhaar i. Name - from doc ii. DOB - from doc iii. State - from doc iv. Photo - from doc v. If QR code can be read, verify whether the user's name from QR source matches with that on doc

- vi. If QR code can't be read, do Aadhaar Lite, verify whether the State & Age matches with that on doc
- b. For DL - Integrate with IDFY api (integration doc - todo), validate user's Name from source with that on doc & store user's following info
 - i. Name - from source
 - ii. DoB - from source
 - iii. State - from source
 - iv. Photo - from source
- c. For Voters - Integrate with IDFY api (integration doc - todo), validate user's Name from source with that on doc & store user's following info
 - i. Name - from source
 - ii. DoB - from source
 - iii. State - from source
 - iv. Photo - from doc
- d. For Passport - Integrate with IDFY api (integration doc - todo), validate user's Name from source with that on doc & store user's following info
 - i. Name - from source
 - ii. DoB - from source
 - iii. State - from doc
 - iv. Photo - from doc

Name Match Scenarios with PAN if verified (Internal algorithm)

- If name match ratio = VERIFIED → Approve
- If name match ratio = IN REVIEW → Move in manual queue (Risk team to action on these and decline if the name is not matching)
- If name match ratio = REJECTED → Decline

6.10

Success & Error states

Success:

- User to be marked as min-KYC Verified if we are able to get the following params:
 - State
 - Name

	◦ Age (DoB) ◦ ID Number Error : • Blocked state : When a user has uploaded a Blocked State document, then we still give them an option to upload a secondary document, but the name of the secondary doc should match with the name as on the primary one. • User is a minor (less than 18 years of age) • Unable to read name • Unable to read state • Unable to read DOB • Unable to read Photo • Unable to validate from source
6.11	If a doc is unlinked, then their Min / Full KYC status to be updated
6.10	RPA of Full-KYC Apart from Min-KYC, verify user's following info : • Obtain user's photo as per ID (this would have come from Min KYC itself) • Liveliness Check (user's selfie should match with their photo on their id card)
6.10	Full-KYC flow can be invoked at the time of & is configurable via a Feature Flag • Add Cash • Paid contest join • Withdrawal • If the life time cumulative deposits for a user is more than Rs. 100k (configurable), user has to verify full-KYC • If sum of all transactions (all deductions / credits in the Deposit & Withdrawal buckets) are >10k in this calendar month • If sum of all transactions (all deductions / credits in the Deposit & Withdrawal buckets) are >120k in this calendar year • Deeplink
7	Visibility for PAN, KYC & PAN+KYC verified users on various touch points

- Leaderboard - check mark on Profile image or Team Name
- Profile - check mark on Profile image or Team Name

Edge cases

Add

- ~~Comms on terminal status for verification if its async. should we do for sync ?~~
- ~~Check for PAN / Bank / UPI / KYC dedupe after we get response from sourcee~~
- Events
- Rollout - DRS
- PAN
 - verification - configurable ?
 - image - configurable ?
- ~~VPA - multiple VPA against same mobile number?~~
- ~~Unlink Bank, UPI or Email - no of times to unlink ?~~
- Bank Verification flow - Sync flow ?
- ~~New touch points on apps to verify users ?~~
 - ~~asking users to verify from home, contests,~~
 - ~~check mark on profile, team name once user is verified~~
- ~~KYC name match instead of NSDL name match for~~
 - ~~Bank verification~~
 - ~~UPI verification~~
- ~~min KYC - Aadhaar number validation logic at BE~~
- Verification timeline / steps ?
- min-KYC?
- Verifications threshold rate limit
 - alerting us if theres any brute force attack
- comms - add transactional sms

- what is min-kyc / full-kyc ?

TBD

- Min-KYC
 - ~~Aadhaar — can we rely on Aadhaar's DOB ?~~
- ~~Dedupe across all KYC modes~~
 - ~~Name match & DOB check~~
- KYC OCR
 - Validating from source
 - Documents are missing for some users
 - ~~wrong 8 digits masked in image~~
 - ~~Some QR code aren't masked~~

Future scope

- Introduce Full KYC

Misc

3.4.1

~~Option 1 — only user inputs with Traces API~~

- Input
 - User entered PAN number, User entered name, User entered DOB
- Flow
 - NSDL call with PAN number, Name, DOB
- NSDL response
 - If invalid PAN, show error
 - If valid PAN, Name not matching, DOB not matching, show error
 - If valid PAN, Name not matching, DOB matching, show error
 - If valid PAN, Name matching, DOB not matching, show error
 - If valid PAN, Name matching, DOB matching, verify PAN

- Handling of false negative
 - Name not matching
 - Get name from Traces API
 - Name match with Traces API name & user entered name
 - If name matches, send NSDL request with traces name, user entered pan number, user entered DOB
 - If name not matches, show error.
 - DOB not matching
 - Get DOB from PAN plus / comprehensive API
 - Show error with the DOB from PAN plus / comprehensive API ???
- Issues with this flow
 - PAN image isn't captured
 - Reliant on other scraped sources for verifying

3.4.2

~~Option 2~~ **only PAN image**

- Input
 - PAN Image
- Flow
 - Validate document if its PAN
 - Check if PAN is not tampered
 - Get PAN number, name, DOB from OCR
 - NSDL call with PAN number, Name, DOB
 - NSDL response
 - If invalid PAN, show error
 - If valid PAN, Name not matching, DOB not matching, show error
 - If valid PAN, Name not matching, DOB matching, show error
 - If valid PAN, Name matching, DOB not matching, show error
 - If valid PAN, Name matching, DOB matching, verify PAN
- Handling of false negative - Manual queue
 - Name not matching
 - Verify PAN if user entered name is exactly matching with PAN image's name
 - DOB not matching

- Verify PAN if user entered DOB is exactly matching with PAN image's DOB
- Issues with this flow
 - TAT will increase due to processing of PAN image for validation, tampering & OCR
 - No way to edit PAN name or PAN DOB

3.4.3

~~Option 3 – PAN image with inputs~~

- Input :
 - User entered PAN number, User entered name, User entered DOB
 - PAN Image
- Flow
 - NSDL call with PAN number, Name, DOB
 - NSDL response
 - If invalid PAN, show error
 - If valid PAN, Name not matching, DOB not matching, show error
 - If valid PAN, Name not matching, DOB matching
 - Validate document if its PAN
 - Check if PAN is not tampered
 - Get PAN number, name, DOB from OCR
 - Verify PAN if OCR details match with user entered details
 - If valid PAN, Name matching, DOB not matching
 - Validate document if its PAN
 - Check if PAN is not tampered
 - Get PAN number, name, DOB from OCR
 - Verify PAN if OCR details match with user entered details
 - If valid PAN, Name matching, DOB matching,
 - Validate document if its PAN
 - Check if PAN is not tampered
 - Get PAN number, name, DOB from OCR
 - Verify PAN if OCR details match with user entered details
- Handling of false negative - handled by OCR details
- Issues with this flow

- User needs to enter PAN details as well as upload the image
- Three way check with - User Input <> PAN image <> NSDL
- TAT will increase due to processing of PAN image for validation, tampering & OCR

3.4.4

~~Option 4 – PAN image (with inputs in case of errors)~~

- Input :
 - PAN Image
 - If first attempt has failed - prefilled from OCR
 - User entered PAN number, User entered name, User entered DOB
- Flow for first attempt
 - Validate document if its PAN
 - Check if PAN is not tampered
 - Get PAN number, name, DOB from OCR
 - NSDL call with PAN number, Name, DOB
 - NSDL response
 - If invalid PAN, show error
 - If valid PAN, Name not matching, DOB not matching, show error
 - If valid PAN, Name not matching, DOB matching, show error
 - If valid PAN, Name matching, DOB not matching, show error
 - If valid PAN, Name matching, DOB matching, verify PAN
- Flow for second attempt onwards
 - NSDL call with PAN number, Name, DOB
 - NSDL response
 - If invalid PAN, show error
 - If valid PAN, Name not matching, DOB not matching, show error
 - If valid PAN, Name not matching, DOB matching, show error
 - If valid PAN, Name matching, DOB not matching, show error
 - If valid PAN, Name matching, DOB matching, verify PAN
 - Validate document if its PAN
 - Check if PAN is not tampered
 - Get PAN number, name, DOB from OCR
 - Verify if details match

- Handling of false negative - Manual queue
 - Name not matching
 - Verify PAN if user entered name is exactly matching with PAN image's name
 - DOB not matching
 - Verify PAN if user entered DOB is exactly matching with PAN image's DOB
- Issues with this flow
 - TAT will increase due to processing of PAN image for validation, tampering & OCR

3.4.5

~~Option 5 – PAN image (with inputs in case of errors)~~

- Input :
 - User entered PAN number, User entered name, User entered DOB
 - If first attempt has failed
 - PAN Image
- Flow for first attempt
 - NSDL call with PAN number, Name, DOB
 - NSDL response
 - If invalid PAN, show error
 - If valid PAN, Name not matching, DOB not matching, show error
 - If valid PAN, Name not matching, DOB matching, show error
 - If valid PAN, Name matching, DOB not matching, show error
 - If valid PAN, Name matching, DOB matching, verify PAN
- Flow for second attempt onwards
 - NSDL call with PAN number, Name, DOB
 - NSDL response
 - If invalid PAN, show error
 - If valid PAN, Name not matching, DOB not matching, show error
 - If valid PAN, Name not matching, DOB matching
 - Validate document if its PAN
 - Check if PAN is not tampered
 - Get PAN number, name, DOB from OCR
 - Verify PAN if OCR details match with user entered details

- If valid PAN, Name matching, DOB not matching
 - Validate document if its PAN
 - Check if PAN is not tampered
 - Get PAN number, name, DOB from OCR
 - Verify PAN if OCR details match with user entered details
- If valid PAN, Name matching, DOB matching
 - Validate document if its PAN
 - Check if PAN is not tampered
 - Get PAN number, name, DOB from OCR
 - Verify PAN if OCR details match with user entered details
- Handling of false negative - handled by OCR details
- Issues with this flow - For the 2nd try
 - User needs to enter PAN details as well as upload the image
 - Three way check with - User Input <> PAN image <> NSDL
 - TAT will increase due to processing of PAN image for validation, tampering & OCR